Lecture 19

Theorem 16. *Let $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ be (column) vectors in $\mathbb{R}^n$ and A be the $n \times n$ matrix obtained by making vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ its columns in respective order, then vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ are linearly independent if and only if $det(A) \neq 0$.*

Proof. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be scalars such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n = \mathbf{0}$ and let $\mathbf{x} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}$ notice that $A\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n$ and so $A\mathbf{x} = \mathbf{0}$ has only one (trivial) solution if and only if A is non-singular if and only if A is invertible if and only if $det(A) \neq 0$. □

Example: Vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 5050 \\ 2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ are linearly independent in usual $\mathbb{R}^3$ since $det \begin{pmatrix} 1 & 5050 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} = 4 \neq 0.$

Basis and dimension

Definition 34. *Let V be a vector space, vectors $v_1, v_2, \ldots, v_n$ are said to form a basis for V if*

1. *$v_1, v_2, \ldots, v_n$ are linearly independent;*
2. *$Span(\{v_1, v_2, \ldots, v_n\}) = V$.*

Examples: (1) Consider $\mathbb{R}^3$ (with usual addition and scalar multiplication)

- Vectors $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$ form a basis for $\mathbb{R}^3$ (called the standard basis for $\mathbb{R}^3$). In previous lecture notes we have seen that those vectors are linearly independent and span$(\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}) = \mathbb{R}^3$.
- Vectors $(1, 0, 0), (5050, 2, 0)$ and $(3, 1, 2)$ also form a basis for $\mathbb{R}^3$. They are linearly independent (see very first example) and to check that they span $\mathbb{R}^3$ we check that for each $(x_1, x_2, x_3) \in \mathbb{R}^3$ there exists λ_1, λ_2 and λ_3 such that $\lambda_1(1, 0, 0) + \lambda_2(5050, 2, 0) + \lambda_3(3, 1, 2) = (x_1, x_2, x_3)$.

$$\left(\begin{array}{ccc|c} 1 & 5050 & 3 & x_1 \\ 0 & 2 & 1 & x_2 \\ 0 & 0 & 2 & x_3 \end{array}\right)$$

  Notice that the above system is already in triangular form and so the solution exists for each $(x_1, x_2, x_3) \in \mathbb{R}^3$.
- Vectors $(1, \pi, 7), (3, 2, 9), (1, 2, 100)$ and $(30, 40, 9)$ do not form a basis for $\mathbb{R}^3$. These are 4 vectors in $\mathbb{R}^3$ and so linearly dependant.

(2) Consider P_2 the set of polynomials of at most degree 2 (with coeffiecient-wise addition and scalar multiplication).
Polynomials $1, x,$ and x^2 form a basis for P_2 (quick to check).

(3) Consider $M_{2\times 2}(\mathbb{R})$ the set of 2×2 matrices (with component-wise addition and scalar multiplication).
Matrices $M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, M_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, M_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $M_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ form a basis for $M_{2\times 2}(\mathbb{R})$ (also quick to check).

Theorem 17. *Let V be a vector space (over F), $v_1, v_2, \ldots, v_k$ vectors in V and S a subspace of V such that Span$(\{v_1, v_2, \ldots, v_k\}) = S$. If vectors $v_1, v_2, \ldots, v_k$ are linearly independent, then $v_1, v_2, \ldots, v_k$ form a basis for S.*

Proof. This follows from the fact that S is a vector space in its own right and $\mathrm{Span}(\{v_1, v_2, \ldots, v_k\}) = S$ with $v_1, v_2, \ldots, v_k$ linearly independent. □

Example: In $\mathbb{R}^3$ (with usual addition and scalar multiplication),
$P = \mathrm{Span}(\{(2,0,1),(1,2,3)\})$ is the plane through the origin containing vectors $(2,0,1)$ and $(1,2,3)$. Note that $(2,0,1)$ is not a scalar multiple of $(1,2,3)$ and so $(2,0,1)$ and $(1,2,3)$ are linearly independent and thus form a basis for P.

Theorem 18. *Let V be a vector space (over F), $v_1, v_2, \ldots, v_n$ a basis for V and $w \in V$, then w can be expressed as a linear combination of $v_1, v_2, \ldots, v_n$ in a unique way.*

Proof. Since $w \in V = \mathrm{Span}(\{v_1, v_2, \ldots, v_n\})$, we have that w is a linear combination of vectors $v_1, v_2, \ldots, v_n$.
Now if $w = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n$ and $w = \mu_1 v_1 + \mu_2 v_2 + \cdots + \mu_n v_n$ (for $\lambda_i, \mu_i \in F$ with $1 \le i \le n$) we have that $\mathbf{0} = (\lambda_1 - \mu_1)v_1 + (\lambda_2 - \mu_2)v_2 + \cdots + (\lambda_n - \mu_n)v_n$ and since vectors $v_1, v_2, \ldots, v_n$ are linearly independent we have that $\lambda_1 - \mu_1 = \lambda_2 - \mu_2 = \ldots = \lambda_n - \mu_n = 0$ and so $\lambda_1 = \mu_1, \lambda_2 = \mu_2, \ldots, \lambda_n = \mu_n$ and so w is a linear combination of $v_1, v_2, \ldots, v_n$ in a unique way. □

Definition 35. *Let V be a vector space (over F) and $v_1, v_2, \ldots, v_n$ a basis for V. If $w = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n$, then the n-tuple $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is called the coordinate vector of w relative to basis $v_1, v_2, \ldots, v_n$.*

Examples: (1) Take usual P_2, then polynomial $p = 5 - x^2$ has coordinate $(5, 0, -1)$ relative to basis $1, x, x^2$.
(2) Take $M_{2\times 2}(\mathbb{R})$, then matrix $A = \begin{pmatrix} 5 & 2 \\ 4 & 2 \end{pmatrix}$ has coordinate $(5,2,4,2)$ relative to basis
$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ but coordinate $(5,2,2,4)$ relative to basis $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.
Notice that the order of basis elements is important.

> Notice that given w and basis $v_1, v_2, \ldots, v_n$, finding coordinate $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ often reduces to solving linear system 'arising' from $w = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n$.

Theorem 19. *Let V be a vector space (over F) and $v_1, v_2, \dots, v_n$ a basis for V, then*

(a) If $u_1, u_2, \dots, u_k$ are vectors such that $k > n$, then $u_1, u_2, \dots, u_k$ are linearly dependent;
(b) No vectors fewer than n can span V.

Proof. See theorem 2.5.13 and proof on pages $130, 131$ and 132 in your course notes (notation and formulation may be slightly different, but read the proof at least once and make sure you understand it), it uses lemma 4 in our previous lecture notes. □

Theorem 20. *Let V be a vector space (over F), if $v_1, v_2, \dots, v_n$ and $v'_1, v'_2, \dots, v'_m$ are both bases for V, then $m = n$.*

Proof. Let $v_1, v_2, \dots, v_n$ and $v'_1, v'_2, \dots, v'_m$ both be bases for V. Suppose $m \neq n$, then either $m > n$ or $m < n$ and so by previous theorem one of the list our vectors will either be linearly dependant or not span V and this is a contradiction since both both form a basis for V. □

The previous theorem tells us that all bases of a vector space V have the same number of elements and so we have the following definition.

Definition 36. *Let V be a vector space (over F) and $v_1, v_2, \dots, v_n$ a basis for V, then by dimension of V we mean the number of basis elements, that is, dim$(V) = n$.*

Examples: (1) Consider usual $\mathbb{R}^n$ then $\dim(\mathbb{R}^n) = n$, use standard basis $(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1)$.
(2) Consider usual P_n then $\dim(P_n) = n + 1$, use basis $1, x, \dots, x^n$.
(3) Consider $M_{m\times n}(\mathbb{R})$ then $\dim(M_{m\times n}(\mathbb{R})) = mn$ (easy but make sure you know why).